

THE IMPACT OF IMMIGRATION ENFORCEMENT ON THE U.S. FARMING SECTOR

GENTI KOSTANDINI, ELTON MYKEREZI, AND CESAR ESCALANTE

We examine the effects of local immigration enforcement efforts on U.S. agriculture in dozens of U.S. counties from 2002–2010 by using variations in the timing of adoption of 287(g) programs, which permit local police to enforce immigration law. Difference-in-differences models using microdata from the American Community Survey (2005–2010 waves) and county tabulations from the Census of Agriculture (1997, 2002, and 2007) yield robust evidence that county enforcement efforts have reduced immigrant presence in adopting jurisdictions. We also find evidence that wages of farm workers, patterns of farm labor use, output choices, and farm profitability may have been affected in a manner consistent with farm labor shortages.

Key words: impacts, immigration enforcement, undocumented workers, agricultural sector.

JEL codes: J61, Q10, Q12.

Agriculture in the United States is highly reliant on immigrant workers (Seid 2006; Levine 2009). Thus, the implications of immigration law for the supply of labor available to U.S. farms has occupied a prominent space in the policy debate for many years.

Immigration enforcement in the United States is, however, evolving rapidly, with states and counties taking a more active role in enforcing immigration law in recent years (Pham and Van 2010). The impact that these decentralized efforts have on the supply of agricultural labor is not yet understood. This study uses county-level variation in enforcement efforts generated by the timing of adoption of immigration enforcement programs under section 287(g) of the Immigration National Act (INA), to empirically examine their implications for agriculture.

The passage of the Illegal Immigration Reform and Immigrant Responsibility Act (IIRIRA) of 1996 added Section 287(g) to the INA. Section 287(g) allows the U.S. Immigration and Customs Enforcement (ICE) deputy director to enter into agreements with state and local authorities,

thereby permitting designated local officers to perform immigration law enforcement functions, provided they receive appropriate training and function under the supervision of ICE officials (Capps et al. 2011).

Even though the IIRIRA was signed into law in 1996, the first contract—between the ICE and the Florida Department of Public Safety—was not executed until 2002. Between 2002 and 2011, 69 U.S. jurisdictions signed an enforcement contract under Section 287(g), including 10 states (e.g., Alabama, Florida, Arizona, and Georgia) and 47 counties (e.g., Cobb County, Georgia; Gaston County, North Carolina; Davidson County, Tennessee).

By 2011, 186,000 illegal immigrants had been identified for removal under this program (Parrado 2011). While the program was predominantly designed to remove illegal immigrants with criminal records, ICE data indicates that about half of the program's activity (defined by the number of immigration detainers issued) has not involved felons, but rather immigrants who were detained for misdemeanors (e.g., open container of alcohol) and traffic offenses (Capps et al. 2011).

Many reports present evidence on the use of 287(g)-granted powers to pursue local immigrant reduction goals, especially among county-led efforts. For example, Weissman (2009) reports that in May 2008, 83% of the

Genti Kostandini is an assistant professor at the University of Georgia, Elton Mykerezzi is an associate professor at the University of Minnesota, and Cesar Escalante is an associate professor at the University of Georgia. The authors want to thank J. Edward Taylor and one anonymous reviewer for their valuable comments. Correspondence may be sent to: gentik@uga.edu.

immigrants arrested in Gaston County, North Carolina were charged with traffic violations. Coleman and Kocher (2011) claim that localizing immigration enforcement amounts to a “spatialized tactic of immigrant incapacitation.” Such reports have led many advocacy groups to voice concerns over the impact of the program on human rights and the daily lives of non-violent immigrants, as well as ethnically diverse U.S. citizens (Lacayo 2010).

There is some case-study evidence that 287(g) programs have succeeded in reducing immigrant population in adopting areas (Capps et al. 2011; Rivas 2008; Barry 2009; Juby and Kaplan 2011). For instance, Frederick County, Maryland, which signed the agreement in 2008, arrested almost 80% of their detainees for petty crimes or traffic offenses, and experienced a 61% drop in their Hispanic population, while neighboring counties experienced an increase (Capps et al. 2011). To date, however, there has been no systematic examination of the effect that local enforcement of 287(g) programs has on immigrant populations.

One previous study by Pham and Van (2010) examined the impact that local immigration enforcement efforts, including county-level 287(g) contracts, had on labor markets. The authors used county-level economic data from the U.S. Census Bureau's County Business Patterns, a dataset that reports county-level tabulations based on non-farm businesses only. These authors find that local enforcement efforts have, on average, had a small negative impact on unemployment, but that these efforts have had heterogeneous impacts by industry, even within high immigrant employing industries. Further, Pham and Van (2010) examine if these effects were induced by changes on the “demand side” (i.e., employers find it less rewarding to hire undocumented workers) or by “supply side” pressures (i.e., immigrants leave high enforcement jurisdictions). These authors find some evidence that the effects are consistent with demand-side factors.

This paper focuses on the impact that program adoptions by county authorities have had on immigrant presence in adopting jurisdictions, on earnings of farm workers, farm input use (including labor, machinery, and fuel), farm output mix, and farm income.

We augment the literature on the implications of immigration enforcement in general by examining the impact of local enforcement on immigrant population

counts. Involving local police in immigration enforcement directly reduces immigrant presence through increased arrests of immigrants, makes it riskier for current immigrants to reside in enforcement areas, and may deter new immigrants from moving into such areas. The impact that such involvement has on immigrant populations is an open question of primary importance, and it provides a direct test for the existence of “supply side” pressures on labor markets.

Additionally, this is the only empirical examination of how 287(g) contracts affect agriculture. The issue of immigration laws and labor shortages in agriculture has been a prominent one for the past decade. Yet the impact of the recent spike in county-level enforcement efforts on agriculture has not been examined. Particularly, the Pham and Van (2010) finding that impacts of local enforcement are heterogeneous across industries implies that implications for agriculture have to be inferred through industry-specific studies.

Local variation in the timing of adopting 287(g) programs provides a quasi-experiment that allows us to assess the implications of these immigration laws for agriculture. Specifically, using individual level data from the American Community Survey (ACS), we first provide evidence on whether 287(g) adoptions result in disproportionate changes in immigrant populations and farm worker wages, using a generalized difference-in-differences (DD) approach. This empirical strategy allows us to examine relative changes (in program vs. control counties) in immigrant presence and wages of farm workers by number of years elapsed from the year of adoption. Falsification tests are also provided by using multiple years of pre-program data.

Similar to Pham and Van (2010), we then use county-level tabulations of farm survey data from the U.S. Census of Agriculture to assess the impact that 287(g) programs have on agricultural labor use, other inputs, crop choices, and farm incomes. This analysis uses simple DD regressions based on the 2002 and 2007 survey waves, and provides falsification tests using pre-program data from the 1997 survey.

The paper proceeds as follows. Section two reviews the literature on immigration and agricultural labor, and provides a description of 287(g) programs. Section three lays out

the methodology and data used in the study. Results are presented in section four and conclusions are drawn in section five.

Immigration Laws and Implications for Agricultural Labor Supply

The effect that immigration laws have on agricultural labor has generated ample discussion, particularly since the passage of the 1996 Immigration Reform and Control Act (IRCA), which legalized around 3 million illegal workers, but imposed new sanctions for knowingly hiring undocumented workers. While industry experts hinted that newly-legalized workers would leave agriculture to move to other sectors, [Tran and Perloff \(2002\)](#) found that this was not the case. These authors found that job transition probabilities (into and out of agriculture) did not change for those legalized soon after IRCA relative to other workers unaffected by reform.

The IRCA also took measures to protect farm labor; it revised the H-2 guest worker program by establishing the H-2A agricultural guest worker program, which has no limit on visas, while the non agricultural H-2B program is subject to an annual limit (of 66,000, as of 2013) ([U.S. Department of Homeland Security 2013](#)).

Options for immigration enforcement continued in the 2000s, with proposals of stricter border control, mandatory status verification (E-Verify), pathways to legal status, and use of guest worker programs being offered by the U.S. Department of Labor. However, no major overhaul took place, despite the apparent agreement among political forces that such is needed. In response, farmer group representatives and the media continued to argue for a “farm labor shortage,” often resulting in high quantities of unharvested products ([Rural Migration News 2006](#); [Preston 2007](#)).

Others argue that the notion of “farm labor shortages” should be defined in terms of shifts in agricultural wages and output. For example, [Martin \(2007\)](#) argues that a shortage would cause sharp increases in agricultural wages and decreases in the output of labor-intensive products, yet the author finds neither. [Levine \(2009\)](#) finds no evidence of a nationwide shortage in farm employment, although shortages in particular areas at specific times may exist. [Holt \(2008\)](#),

however, argues that the right metric in an open economy is the United States’ share of world markets in labor-intensive crops, and argues that by this measure, farm labor would appear to have become relatively scarce in the early 2000s.

In more recent years, there has been a decline in the foreign labor supply to U.S. farms, and this trend is likely to continue, in part due to conditions in Mexico ([Taylor, Charlton, and Yunez-Naude 2012](#)). [Hertz and Zahniser \(2012\)](#) note that the number of immigrants dropped by nearly 13% between 2007 and 2011, and find shifts in farm wages consistent with this reduction over the same period.

While much of the empirical evidence on immigration laws and agricultural labor comes from studies that are mostly descriptive in nature, others use simulations of various policy scenarios. For instance, [Devadoss and Luckstead \(2011\)](#) use theoretical models of U.S. and Mexican agriculture and simulation analysis to predict the effect that more restrictive immigration laws would have on agricultural wages and employment.

[Zahniser et al. \(2012\)](#) employ a computable general equilibrium model, and suggest that a gradual annual increase in the number of H-2A workers would result in a 1-2% increase in agricultural output for labor-intensive sectors associated with a decrease of 6% in U.S.-born agricultural workers, and a decrease (3%) in agricultural wages. In addition, tighter immigration laws may result in a 2-4% relative decrease in the output of labor-intensive sectors and a 3% relative increase in the earnings of legal workers.

The passage of IRRA in 1996, followed by more recent increases in state and county enforcement efforts, pose new questions on their impact. Many organizations and researchers claim that 287(g) agreements have created an insecure environment for immigrants, as well as the Hispanic population in general ([Immigration Policy Center 2012](#); [Lopez and Minushkin 2008](#); [Quereshi 2010](#)). For example, a survey from the Pew Hispanic Center found that one in ten Hispanic adults report that they have been asked about their paperwork by authorities ([Lopez and Minushkin 2008](#)). In addition to the direct removal of detained immigrants, the new-found insecurity is likely to drive current immigrant populations away from high enforcement areas, and discourage new

migration in such areas. However, evidence of such an effect on immigrant populations remains largely anecdotal (Weissman 2009; Lacayo 2010; Juby and Kaplan 2011).

What little evidence of an association between local enforcement and agricultural labor exists is also anecdotal or based on small surveys and case studies. McKissick and Kane (2011), for example, conducted a survey among fruit and vegetable farmers in Georgia (which implemented strict statewide immigration enforcement laws in 2011), and found that farmers reported up to a 40% reduction in labor availability compared to the normal peak harvest employment. However, reports of “labor shortages” are not unique to areas that are subject to increased enforcement. For example, Hotakainen (2011) reports that apple growers in the U.S. state of Washington also experienced labor shortages in 2011. We examine the consequences of local enforcement by comparing changes in conditions in jurisdictions that step up enforcement relative to others that do not.

Methodology and Data

Overview

Enforcement efforts based on 287(g) agreements are expected to make localities less attractive for undocumented alien workers. Due to fear of being deported, undocumented immigrants may leave these jurisdictions and move to jurisdictions that are not subject to such laws. Additionally, new immigrants may also be deterred from moving to reside or work in jurisdictions with 287(g) programs, as they face a higher risk of being deported. We first test these propositions using individual-level data from the ACS. As legal status is not observed, we instead examine whether the rates of those who report themselves as noncitizens, foreign-born individuals, or foreign-born individuals who have been in the country for 20 years or fewer declines disproportionately in adopting counties.

Given that more than 50% of agricultural workers in the United States are illegal, such movements may significantly affect the supply of agricultural labor. Reductions in the supply of agricultural labor would put upward pressure on farm wages. We would thus expect disproportionate increases in

the wages of those working in agriculture in adopting jurisdictions relative to other communities that did not change their immigration enforcement efforts. This proposition is tested with individual-level data on over 50,000 workers who declare farm labor as their primary occupation in the ACS.

When faced with higher prices, farm owners could adopt less labor-intensive practices and substitute for capital and machinery. We examine possible changes in agricultural input use and crop choices with data from the Census of Agriculture. Unfortunately, the public-use data does not report hours of labor used, or the hourly cost of labor. Rather, only the number of workers and total expenses on labor are reported. This makes a direct examination of labor use and wages impossible. However, we examine changes in the expenditure shares for labor, fuel, and machinery. If labor shortages have placed upward pressure on wages, under fairly general assumptions regarding the agricultural production function, one would expect the cost share of labor to have declined. We also provide a more direct test of whether farmers have changed production methods in response to 287(g) programs by examining the impact on the share of cropland used for vegetable production, a relatively labor-intensive activity (Carroll, Saltz, and Gabbard 2009).

The paper also examines impacts on expense per worker hired and number of workers hired per year. With no information on hours worked, both measures are rather imperfect. A reduction in workers and an increase in expense per worker could proxy for a reduction in labor input and an increase in price, but it could also be because of less worker turnover (with each worker used for more hours during the year). In any case, less turnover is also consistent with a tighter labor market. Finally, we examine whether adopting counties have experienced disproportionate changes in farm income. The profit function is non-increasing in input prices, so farm incomes are expected to have declined disproportionately in 287(g) counties.

Data

This study uses individual-level data from the U.S. Census Bureau’s American Community Survey (ACS) (Ruggles et al. 2010), as well as county-level data from the U.S. Census of Agriculture conducted by USDA’s

Table 1. Counts of Counties and Workers by 287(g) Status (ACS)

	Control	Adopted County	Adjacent	Adopted State	Total
County counts by year					
2005	1,015	1	2	68	1,086
2006	1,005	5	8	68	1,086
2007	895	13	35	143	1,086
2008	761	31	66	228	1,086
2009	734	33	84	235	1,086
2010	723	39	89	235	1,086
County-years	5,134	122	284	977	6,517
Farm worker counts by year					
2005	7,836	65	304	477	8,682
2006	8,087	182	402	592	9,263
2007	7,536	232	510	955	9,233
2008	6,500	343	638	1,495	8,976
2009	6,732	377	680	1,538	9,327
2010	6,778	559	814	1,479	9,630
Worker-years	43,469	1,758	3,348	6,536	55,111

National Agricultural Statistical Survey (NASS) (USDA NASS 2012).

The ACS is a repeated cross section released annually, which collects data on 250,000 households each month. In addition to demographic information such as gender, race, and ethnicity, the survey also elicits information on education and work activities (labor force and employment status, annual earnings, weeks worked, and average number of hours worked per week). Most importantly, the occupation of the respondents' primary job (defined as the one that takes up the majority of one's weekly time) is elicited. This makes it possible to focus attention on the earnings of farm workers.

Immigration information is also elicited; respondents report their place of birth, and if it is outside of the United States, they report the number of years that they have lived in the country, and whether they are U.S. citizens (including permanent residents).

Finally, the ACS provides geographic information at levels of aggregation below the state level, but the county where respondents reside is not always reported. Starting with the 2005 wave, the ACS reported counties if they exceed a population of 65,000, and has created geographic areas called Public Use Micro Data Areas (PUMAs) that are available for all respondents. The PUMAs are areas with a minimum population of 100,000 people, and they do not cross state lines. Larger counties contain several PUMAs, while multiple smaller counties may be

contained within PUMAs. The vast majority of counties are either fully contained in a PUMA or fully contain several PUMAs.

The 287(g) programs are implemented at the county level. Thus, for individuals in PUMAs that contain multiple counties and at least one program county, it is not possible to determine with certainty whether they are subject to the program or not. Therefore, we exclude PUMAs with multiple counties and at least one program county from the analysis. All PUMAs that are fully contained in a county, or those that do not contain a program county, are used. This excludes 10 of the 49 program counties from the analysis that uses ACS data. The remaining areas include either identifiable counties (with or without a program), or multi-county PUMAs known not to have a program.

Table 1 presents the total number of areas used in the analysis, the number of jurisdictions that adopted each year, as well as the number of farm workers in each area by survey year. Summary statistics for the variables used from the ACS by type of jurisdiction are presented in table 2 for the first survey year (2005) to provide a view of baseline differences between would-be adopters and control areas.

In addition to the individual-level data, this study also utilizes county-level tabulations from the Census of Agriculture (1997, 2002, and 2007) for all U.S. counties. The latter is a mandatory census of all farmers conducted every five years from the USDA NASS.

Table 2. Summary Statistics by 287(g) Status (2005 ACS)

	Control		Adopted County		Adjacent		Adopted State	
	Mean	S.E	Mean	S.E	Mean	S.E	Mean	S.E
Log earnings	9.095	1.289	9.286	1.083	9.245	1.127	9.156	1.243
Not citizen	0.398	0.489	0.567	0.496	0.613	0.487	0.373	0.484
Demographics								
Age	32.972	12.604	34.58	11.543	35.215	11.823	33.346	12.248
Female	0.253	0.435	0.253	0.435	0.282	0.45	0.241	0.428
Black	0.031	0.173	0.033	0.178	0.041	0.198	0.081	0.273
Hispanic	0.455	0.498	0.661	0.474	0.649	0.478	0.369	0.483
High school	0.317	0.465	0.29	0.455	0.274	0.446	0.35	0.477
Some college	0.098	0.297	0.053	0.225	0.075	0.263	0.097	0.296
College or better	0.005	0.068	0.002	0.049	0.004	0.06	0.003	0.054
Weeks worked								
0–14	0.183	0.387	0.126	0.332	0.145	0.353	0.151	0.358
14–26	0.125	0.331	0.169	0.375	0.148	0.355	0.102	0.303
27–39	0.101	0.301	0.077	0.268	0.103	0.304	0.087	0.281
40–47	0.08	0.272	0.092	0.289	0.103	0.304	0.057	0.232
48–49	0.03	0.17	0.011	0.104	0.052	0.221	0.027	0.161
50–52	0.48	0.5	0.525	0.5	0.449	0.498	0.577	0.494
Hours per week	41.584	15.591	40.207	14.322	40.37	13.478	41.144	14.679
N	6,167		328		704		1,483	

County-level information on labor expenses, number of hired workers, fuel expenses, value of machinery, total land in farms and cropland, acres planted with vegetables, and farm income is provided, among other variables.

Information on the implementation of 287(g) programs by state and county was collected from the ICE (United States Immigration and Customs Enforcement 2011). Information on visible immigration raids was also collected through an extensive search of news articles and other printed media. While the analysis focuses on 287(g) programs at the county level, state efforts and incidence of raids are controlled for in the multivariate analysis.

Descriptive statistics based on the county-level data for all the variables used in this study for year 2002 (the first pre adoption year) are shown in table 3, and information on counties and states that adopted 287(g) programs is presented in table A1 in the appendix. Statistics are presented separately for all program counties, for counties that are adjacent to a program area but not themselves subject to enforcement, as well as all other areas that have neither enacted a program nor are adjacent to one.¹

Empirical Model

We first examine whether county-level contracts between ICE and local law enforcement agencies have exerted an effect on: (a) the incidence of self-reported noncitizens; (b) immigrants in general; and (c) immigrants who have spent 20 years or fewer in the United States. We then examine the impact on earnings exercised by workers who are not self-employed and report farm work as their primary occupation. A DD approach is used to explain variation in individual outcomes within a county over time. Specifically, for each individual i that resides in county c and state s in year t , the following equation is estimated:

$$(1) \quad Y_{icst} = \theta Z_{ct} + \beta_1 S_{st} + \beta_2 R_{ct} + \alpha X_{icst} + \gamma_c + \tau_t + \varepsilon_{icst}.$$

Depending on the specification, Y_{icst} measures earnings or immigrant status for individual i in county c , state s and year t . A measure of the county's program status in year t , (Z_{ct}) is included to capture program effects. Since states also adopted 287(g) programs, instead of treating counties in such

¹ If a county signed a 287(g) agreement and is also part of a state that also signed an agreement, it is coded as a program county. Also, any adjacent areas that also implement a program

or are in states that have implemented a program are coded as county-level and state-level program areas as opposed to adjacents.

Table 3. Summary Statistics by 287(g) Status, 2002

	All Counties			Adopted County			Adjacent County			Control		
	Mean	S.E.	N	Mean	S.E.	N	Mean	S.E.	N	Mean	S.E.	N
Hired labor share	0.101	0.077	2,999	0.183	0.116	17	0.103	0.082	54	0.098	0.075	2,435
Expense per worker	4,879	2,857	2,995	8,007	2,615	17	4,453	2,718	54	4,718	2,719	2,431
Workers per acre	0.876	6.215	3,042	2.087	3.282	19	0.788	1.291	56	0.843	6.261	2,461
Share of fuel expenses	0.049	0.02	3,046	0.035	0.012	19	0.044	0.017	56	0.05	0.02	2,462
Log of machinery value	5.054	0.94	3,056	5.657	0.501	19	5.303	0.629	56	5.095	0.885	2,469
Share of vegetable acres	0.008	0.022	2,134	0.009	0.02	19	0.007	0.022	50	0.007	0.019	1,716
Log of farm income per acre	3.448	1.481	2,618	4.794	1.529	18	3.858	1.782	45	3.413	1.374	2,130
Land in 100k acres	3.035	3.894	3,064	1.974	2.233	19	2.368	3.775	57	3.008	3.354	2,475
Cropland in 100k acres	1.413	1.49	3,069	0.72	0.728	19	0.989	1.403	57	1.551	1.503	2,478

states as controls, we explicitly include indicators for counties in states and years with active 287(g) contracts at the state level (S_{st}). As noted in [Pham and Van \(2010\)](#), substantial policy change occurs at the state level each year, making it difficult to isolate state-level enforcement effects. We thus interpret the state-level program estimates as controls, and do not attempt to infer a causal relationship.

In addition to 287(g) programs, the ICE has also intensified the number of visible, large-scale immigration raids to apprehend undocumented immigrants. The number of undocumented immigrants arrested at workplaces increased more than seven-fold, from 500 to 3,600 between 2002 and 2006 ([Capps et al. 2007](#)). But more importantly, work site raids targeted and removed hundreds of immigrants on the same locality, and brought high profile criminal (fraud) charges to those who were found to be working without proper documentation. We conducted a web search for instances of such raids that were reported in the press and added controls for these (R_{ct}).

Finally, in the earnings equations, we also control for individual-level observable influences on wages (X_{icst}), including indicators of gender, race, and ethnicity. Indicators of education level are also included, as are age and age squared (to proxy for experience)

and time worked over the reference period (number of weeks and average hours per week).

Unobservable influences are partitioned into three additive components: a county fixed effect captures time-stable county-specific influences (γ_c), a year fixed effect (τ_t) that captures differences in outcomes over all individuals across years, and idiosyncratic influences remain in the error term (ϵ_{icst}). To allow θ to capture the average effect of enforcement, we first define Z_{ct} as a post adoption indicator $1(\text{Law}_{ct})$ for programs adopted at the county-level in post adoption years.

Because 287(g) participation is not randomly assigned, there may be systematic unobserved differences between counties that influence both program adoption and outcomes, which could bias estimates of the program effect. Including county fixed effects allows us to identify program effects by examining within-county changes in outcomes across adopters and control counties. Year fixed effects identify aggregate year-specific effects (τ_t), across all individuals. This generalized DD model allows us to identify calendar year effects and program effects separately because of variation in the timing of adoption across jurisdictions. The model is identified by assuming that

program participation (Z_{ct}) is uncorrelated with time-varying unobserved influences (ε_{ict}) conditional on other observables and fixed effects:

$$(2) \quad \text{Cov}[Z_{ct}, \varepsilon_{ict} | X_{ict}, \gamma_c, \tau_t] \equiv 0.$$

Thus, 287(g) program effect estimates are unbiased, even if program adoption is based on time-stable unobserved differences. For instance, if less immigrant-friendly counties are more likely to adopt or to adopt earlier, this does not bias estimates. However, if areas with growing anti-immigrant sentiment were more likely to adopt, this could cause bias. The crucial assumption is that within county, time-varying, unobserved influences on outcomes are not systematically related to adoption. [Bertrand, Duflo, and Mullainathan \(2004\)](#) stress the importance of choosing the right control group and considering trends leading up to the time of program adoption. The assumption of “no time varying unobservable differences” between the treatment and control group implies that these two groups would follow similar trends in the absence of treatment.

This assumption is testable for trends preceding adoptions, since data is available for multiple years. We test for the presence of differential trends in outcomes among non-adopters and future adopters *prior to adoption*. This is accomplished by measuring program participation status (Z_{ct}) via a full set of leads and lags. We set all county-years in non program (control) counties to zero, as well as the year immediately prior to adoption for adopting jurisdictions. Each remaining year in adopting counties is then specified in terms of its temporal distance from the adoption year (e.g., 1(3 years pre-adoption), 1(2 years pre-adoption), 1(year of adoption), 1(first year post-adoption), and so on).

These leads and lags nonparametrically describe the differences in trends (measured as deviations in each year relative to the last year before adoption) prior to and after the 287(g) start date in adopters relative to the average change in non-adopters over the same number of years.

This makes it possible to determine if adopters and non-adopters were experiencing differential trends in each year leading up to adoption, and if they continue to do so after the program starts. Statistically identical

pre-trends provide evidence that the crucial “parallel trends” assumption is satisfied. We note that this strategy, in addition to testing for pre-trends, also relaxes the constant program effect assumption that is implicit in the first specification, thus allowing the effects to vary by year since program adoption.

Additionally, we identify counties that are adjacent to program counties but are not themselves subject to enforcement, and add leads and lags indicating temporal distance from the time that the first neighboring county adopted a contract. Neighboring areas are of interest for two distinct reasons. First, they are likely to share similar time varying unobservables, so it is more likely that the assumption of parallel trends in the absence of treatment holds. However, program adoption in treatment areas may induce spillover change in adjacent areas. This possibility makes the interpretation of statistical differences in post program trends between adopters and adjacents more complicated.

The strategy of using geographically adjacent counties as controls to identify program effects with DD models has been widely employed in other contexts. For example, studies of minimum wage impacts, such as [Card and Krueger \(1994\)](#), use neighboring areas as controls in a DD approach. Most studies have not explicitly tested for the possibility of spillover effects. One recent study focusing on minimum wage law impacts uses matched pairs of geographic adjacents as treatment and control areas, but also explicitly tests for spillover effects by testing program and adjacent areas against non program, non adjacents ([Dube, Lester, and Reich 2010](#)).

In our case, focusing on adjacent areas can provide a robustness check for the main specification, but also allows us to examine possible spillover effects. If immigrants who move away from (or refrain from locating in) program counties have a higher propensity of ending up in adjacent counties, then one would expect the program to have the opposite effect on adjacent counties.

Thus, we interpret the pre-trend tests between adjacent and program counties as an additional falsification test. If trends in outcomes are parallel between treatment and adjacent areas up until program adoption, one is reassured that assuming that these trends would have continued to be parallel in the absence of the program is reasonable.

Differences in post program trends between program and adjacent counties could measure combined program and spillover effects. However, if non adjacent non program counties also show parallel pre-trends and provide a reasonable control group, comparing adjacent to non adjacent non program areas can provide a direct test for spillover effects, and these can be identified separately from the main program effect on adopters.

The remainder of the analysis examines enforcement effects on agricultural labor use and payments, fuel expenses, crop choices, and farm income with county-level data from 1997, 2002, and 2007. We are still able to estimate DD models to isolate the impact of enforcement, but only two observations for each county can be used to examine program effects since all adoptions occurred after 2002. Thus, we only estimate a typical DD model similar to Card and Krueger (1994), and cannot separate time from program effects. As above, pre-to-post-law differences in outcomes among program counties are compared against outcome differences in two different county groups—all non-adopters, and those adjacent to adopters. Specifically, we estimate:

$$(3) \quad Y_{cst} = \tau T_{07} + \alpha Z_c * T_{07} + \beta_1 P_s * T_{07} \\ + \beta_2 R_c * T_{07} + \gamma_c + \varepsilon_{cst}$$

where Y_{cst} is the outcome variable in county c , state s and year t (e.g., labor use, number of workers hired, labor expense, etc). As noted, all adoptions examined here occurred between 2002 and 2007, so T_{07} indicates the last survey year (after all examined adoptions have taken place). The interaction $Z_c \times T_{07}$ is equal to one for program counties after adoption. Interactions between an indicator of a county in a state subject to a 287(g) program and the indicator of the 2007 survey round ($P_s \times T_{07}$), as well as interactions between a county subject to a raid and the 2007 survey round ($R_c \times T_{07}$) are also included. For the same reasons as above, we also distinguish between adjacent counties and all other counties in this analysis.

In this analysis, omitted time-varying county attributes that are correlated with policy adoption and the outcome Y_{cst} can still bias estimates, but differences in time-stable unobservables are not of concern. Thus, tests

of pre-trends are important to provide evidence on the parallel trends assumption. Here, however, we cannot separate calendar year from program start. Thus, we only provide some evidence on pre-trends by using the 1997 survey. Changes in outcome variables between 1997 and 2002 (last five years before the first program was adopted) are compared for the would-be program counties, would-be adjacents, and counties that do not adopt a program and are not adjacent to a program county.

Additionally, a specification that only uses counties of similar size to adopters and another that adds controls for land use (to examine if any effects are the result of a technology change or just downsizing of production) are estimated.

We focus our attention on the following outcome variables: share of hired labor expense out of total production expense, expense per hired worker, number of workers hired per unit of land, share of fuel expense out of total production expense, machinery value per unit of land, share of vegetable acres per unit of land, and farm income per unit of land. All these variables are available at the county level for 1997, 2002, and 2007.²

Results

Individual-level Analysis

We first gain some insight on the impact of immigration enforcement on immigrant presence by examining pre-to-post adoption changes in the incidence of noncitizens in adopting counties relative to changes in control areas. Table 4 presents results of DD estimates for three outcome variables: the incidence of noncitizens, all foreign-born individuals, and foreign-born individuals who have been in the country for fewer than 20 years.

Results show that 287(g) contracts executed by county-level authorities have a negative impact on the presence of noncitizens in adopting counties, suggesting that

² It has to be noted that hired labor (comprising on average more than 80% of the total farm labor) is just one component of the total farm labor hired by U.S. farms. The other part is comprised of contract labor, which is more seasonal and is brought by contractors and other agricultural intermediaries (Martin 2007). All models are weighted by land in farms in 2002, with the exception of the one for expense per worker (which is weighted by number of workers).

Table 4. Impact of the 287(g) Program on the Share of Immigrants in Area

	Noncitizens	Foreign-Born	Foreign-Born with Less Than 20 Years in U.S.
Post-county	−.0042** (0.0017)	−.0022 (0.0018)	−.0047*** (0.0018)
Post-adjacent county	0.0009 (0.0012)	−.0012 (0.0014)	−.0019 (0.0013)
Post-state	0.0019** (0.0009)	0.0017* (0.001)	0.0036*** (0.001)
Post-raid	−.0058 (0.0048)	−.0045 (0.0091)	−.0094** (0.0046)
N	10,567,024	10,567,024	10,567,024
R ²	0.0748	0.1362	0.0817
Areas	1,086	1,086	1,086

Notes: Standard errors are in parentheses, and *, **, and *** denote statistical significance at the 10%, 5% and 1% levels, respectively. Standard errors are clustered at the PUMA level.

immigration enforcement drives noncitizens to move away from adopting counties, or discourages new immigrants from moving to a program county.

Citizenship status is, however, self reported. So there is a possibility that illegal immigrants in enforcement areas are less likely to self report noncitizen status than otherwise similar peers. Immigrant status (regardless of legal status) may be less likely to be misreported since language differences are likely to be apparent for foreign-born individuals. We thus estimate an additional specification with incidence of any foreign-born individual as a dependent variable. The estimate is negative, but smaller (−0.0022) and not statistically significant at conventional levels.

However, the majority of foreign-born individuals are legal and not affected by the policy. To narrow the analysis down to foreign-born individuals who are more likely to not have legal status, we estimate the impact of enforcement on the incidence of foreign-born individuals who report having been in the United States for fewer than 20 years. The overwhelming majority of immigrants with more than 20 years in the United States are very likely to have obtained legal status (many through IRCA). Estimates indicate a drop of 0.047 ($p < 0.01$), a very similar result to that presented in the first specification.

The “parallel trends” assumption is perhaps the most important in a DD framework. We examine the presence of pre-trends by estimating dynamic models (with full leads and lags relative to the pre-adoption year) for county-level 287(g) programs to examine if the negative effect may in part be due to a pre-existing differential trend. While unlikely, it could be that in the pre-adoption period, the share of noncitizens was actually declining faster in the would-be adopters than in control counties. In that case, the result may simply be the continuation of a trend.

Table 5 presents parameter estimates associated with each pre-adoption year, the year of adoption, and each post adoption year, with the single-year prior to adoption (1 Year Pre) and all non-adopting county-years set to zero. Figure 1 plots the DD estimates and 95% confidence intervals. For each pre-law period, the deviation relative to the pre-adoption year is statistically equal to changes in the control counties, indicating no pre-trends. The hypothesis that the deviations from the pre-adoption year for all years before adoption are jointly equal to zero is tested via F-tests, and results are presented at the bottom of table 5. F-tests do not reject the null that pre-trends are jointly equal to zero.

The difference in immigrant presence during the year of adoption, compared to one year before in adopting counties is equal to the average change in control counties. This is not surprising, since many programs are enacted in the latter half of the year, after nearly half of the ACS sample has been interviewed. In each post-adoption year, the estimates are negative, likely reflecting a true program effect. Most year estimates, as well as all post-law estimates as a group, are statistically different from zero ($p < 0.01$).

Results in table 5 show that the policy produces a constant annual decline in the presence of noncitizens. The estimate in year one after adoption shows a −0.47 percentage point difference relative to the pre-adoption year; by year three the estimate is at −1.1 percentage points, and in year 4 it is −2 percentage points. The average share of noncitizens in adopting counties is 16.1%, so in early years, the incidence of immigrants would be about 3% lower than in pre-adoption years. Four years into the program, immigrant presence relative to the pre-program year is nearly 13% lower.

Table 5. Dynamics of the Impact of the 287(g) Program on the Share of Immigrants in Area

	Noncitizens	Foreign-Born	Foreign-Born with Less Than 20 Years in U.S.
5 years prior	0.0003 (0.0045)	−.0002 (0.0054)	0.0068 (0.006)
4 years prior	0.0011 (0.0043)	−.0028 (0.0048)	0.0047 (0.0043)
3 years prior	0.0042 (0.0027)	0.0043 (0.003)	0.0069** (0.0029)
2 years prior	0.0041* (0.0022)	0.0023 (0.0025)	0.0046** (0.0022)
1 year prior	0.0000 (—)	0.0000 (—)	0.0000 (—)
Year of adoption	0.0006 (0.0022)	0.0017 (0.0024)	0.0019 (0.0022)
1 year post	−.0047** (0.0022)	−.0029 (0.0025)	−.0036 (0.0023)
2 years post	−.0037 (0.0024)	−.0027 (0.0026)	−.0055** (0.0026)
3 years post	−.0111*** (0.003)	−.0079** (0.0032)	−.0153*** (0.0031)
4+ years post	−.0207*** (0.003)	−.0186*** (0.0034)	−.0330*** (0.0033)
5 years pre-adj	0.0006 (0.0049)	0.0019 (0.0051)	0.0003 (0.005)
4 years pre-adj	0.0055* (0.0033)	0.0037 (0.0038)	0.0065 (0.0041)
3 years pre-adj	−.0018 (0.0023)	−.0003 (0.0026)	0.0003 (0.0025)
2 years pre-adj	0.0002 (0.0015)	0.0011 (0.0018)	0.0008 (0.0017)
1 year pre-adj	0.0000 (—)	0.0000 (—)	0.0000 (—)
Year of adoption adj	−.0006 (0.0016)	−.0013 (0.0017)	−.0022 (0.0017)
1 year post-adj	0.0013 (0.0016)	−.0011 (0.0019)	−.0028 (0.0018)
2 years post-adj	−.0007 (0.002)	−.0029 (0.0023)	−.0040* (0.0021)
3 years post-adj	−.0017 (0.0024)	−.0016 (0.0027)	−.0042 (0.0026)
4+ years post-adj	−.0028 (0.0029)	−.0015 (0.0037)	−.0109*** (0.0032)
N	10,567,024	10,567,024	10,567,024
R ²	0.0749	0.1362	0.0818
Tests on leads and lags			
P-value	0.230	0.372	0.130
$(H_0: 1 \text{ Year pre} \dots 5 \text{ Years pre} = 0)$			
P-Value	0.000	0.000	0.000
$(H_0: 1 \text{ Year post} \dots 4 + \text{ Years post} = 0)$			

A decrease in the share of noncitizens following adoption of 287(g) contracts suggests that these programs may reduce labor supply. Next, we use information on the earnings of agricultural workers within the ACS to examine if wages respond to 287(g) adoptions.

Table 6 provides DD estimates of 287(g) adoption on the earnings of farm workers, after controlling for individual characteristics hypothesized to affect earnings, weeks worked over the reference period, and average hours worked per week. The

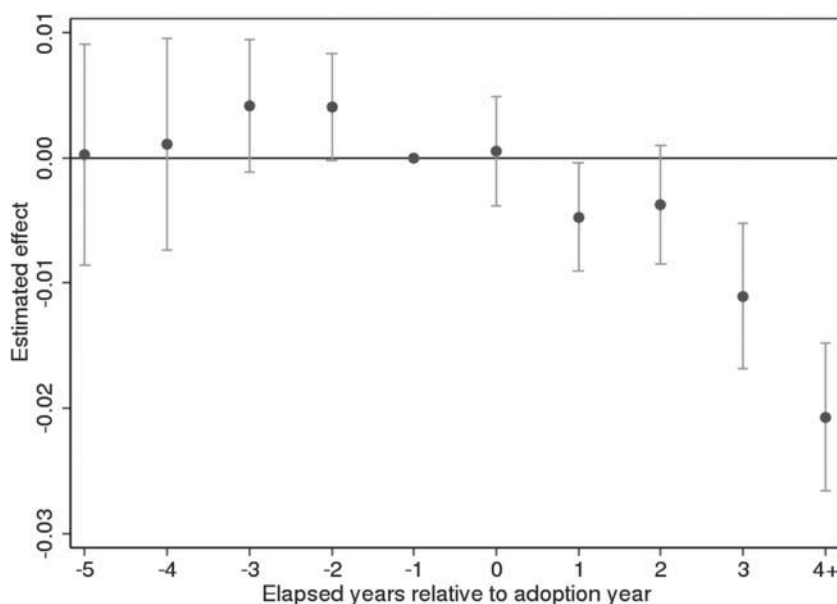


Figure 1. Estimated impact of 287(g) programs on the share of noncitizens for years before, during, and after adoption

dependent variable is the log of earnings from employment in farm entities not owned by the individuals themselves or family members. In the first column, we include indicators of post-287(g) county-years, county-years in areas that are adjacent to an enforcement county after the first neighbor adopted a contract, and control for state-level 287(g) adoption and post-raid county-years. The control group includes farm workers in all counties in the United States that are not subject to 287(g) immigration laws, are not adjacent to such counties, and have not experienced an immigration raid.

Estimates suggest that farm workers in adopting counties may have experienced an increase in wages, but the coefficient is not significant at conventional levels. Farm worker wages in adjacent counties in post-program years, however, have declined disproportionately, and while this difference is not statistically different from wage changes in the control group, it is statistically different from the estimate for post-program county-years.

For purposes of illustration, we estimate a DD model that only compares post-program earning differences between program and adjacent counties. Estimates indicate a 10.8% increase in wages in post-law county-years, relative to same-year changes in non program adjacent counties.

We next turn our attention to the dynamic specifications of each model (with all workers and adjacents only) to examine pre-trends, as well as year-specific post-program effects. The full leads and lags specification with the whole sample, presented in the first column of table 7, indicates no statistically significant pre-trends (individually or as a group), virtually no difference in the year of adoption (consistent with the effect on immigrant presence), and higher relative wages in each ensuing year. These post-program differences are only significant in the first post-adoption year.

When only compared to adjacents, post-program earning changes are all positive, and significant in three of the four post adoption periods and as a group ($p < 0.1$). Pre-trends as a group are not statistically different among all three geographic areas (program counties, adjacents, and non-adjacent controls).

As noted, statistically similar pre-trends indicate that one can reasonably assume that wage paths would have continued to be parallel in the absence of the program. Post-program differences between adopters and adjacents admit two different explanations; they either reflect a true program effect measured against a “better” control group, or the combined influence of a program and a spillover effect. On the other hand, when compared to non-adjacent controls, estimates

Table 6. Impact of the 287(g) Program on Log Earnings

	All Counties	Adjacents
Post-county	0.0672 (0.0473)	0.1082** (0.0499)
Post-adjacent county	-.0620 (0.038)	
Post-state	0.0742*** (0.0284)	0.1196** (0.0499)
Post-raid	-.0171 (0.0595)	
Age	0.0668*** (0.0027)	0.0563*** (0.0064)
Age squared	-.0007*** (0.00003)	-.0006*** (0.00008)
Female	-.1723*** (0.0107)	-.1408*** (0.0236)
Black	-.1159*** (0.0263)	-.1143* (0.0619)
Highest degree high school	0.1072*** (0.0092)	0.1239*** (0.0204)
Some college no BS	0.1704*** (0.0143)	0.1174*** (0.0363)
BS degree or higher	0.1975*** (0.0657)	0.1449 (0.1788)
Worked 14-26 weeks	1.0966*** (0.0213)	1.1337*** (0.0589)
Worked 27-39 weeks	1.5265*** (0.0189)	1.5128*** (0.0489)
Worked 40-47 weeks	1.8134*** (0.0209)	1.7864*** (0.0514)
Worked 48-49 weeks	1.9512*** (0.0259)	1.9398*** (0.0576)
Worked 50-52 weeks	2.0673*** (0.0176)	2.0632*** (0.0447)
Avg. hours worked	0.0223*** (0.0004)	0.0227*** (0.0011)
N	55,111	8,148
R ²	0.6941	0.6731
Areas	1,080	163

for program counties would only reflect a program effect, while those for adjacent counties would only reflect a spillover effect.

This three-way comparison can be more clearly seen in figure 2, where the coefficients and 95% confidence intervals are plotted for program and adjacent counties. Neither

Table 7. Dynamics of the Impact of the 287(g) Program on Log Earnings

	All Counties	Adjacents
5 years pre	-.0179 (0.1408)	-.0732 (0.1373)
4 years pre	0.0113 (0.1067)	-.0205 (0.1119)
3 years pre	0.2063** (0.0917)	0.1939** (0.0932)
2 years pre	0.0271 (0.0845)	0.0324 (0.0872)
1 year pre	0.0000 (—)	0.0000 (—)
Year of adoption	0.0897 (0.0632)	0.0978 (0.0637)
1 year post	0.1493** (0.0758)	0.1866** (0.0769)
2 years post	0.1005 (0.0651)	0.1386** (0.0685)
3 years post	0.1181 (0.0786)	0.1707** (0.0806)
4+ years post	0.0557 (0.0604)	0.0745 (0.0651)
5 years pre adj	-.0914 (0.2042)	
4 years pre adj	0.1722* (0.1044)	
3 years pre adj	0.0608 (0.0575)	
2 years pre adj	-.0443 (0.061)	
1 year pre adj	0.0000 (—)	
Year of adoption adj	-.0601 (0.0512)	
1 year post adj	-.0467 (0.0574)	
2 years post adj	-.0674 (0.0648)	
3 years post adj	-.0978* (0.0511)	
4+ years post adj	-.0375 (0.0507)	
N	55,111	8,148
R ²	0.6941	0.6733
Areas	1,080	163
Tests on leads and lags		
P-value (H ₀ :1 Year pre. . . .5 Years pre = 0)	0.170	0.180
(H ₀ :1 Year pre. . . .4+ Years post = 0)	0.276	0.080

program counties nor adjacents show earnings paths that are statistically different from that in control counties before or after adoption. Thus, we cannot decisively conclude that there is a program or a spillover effect

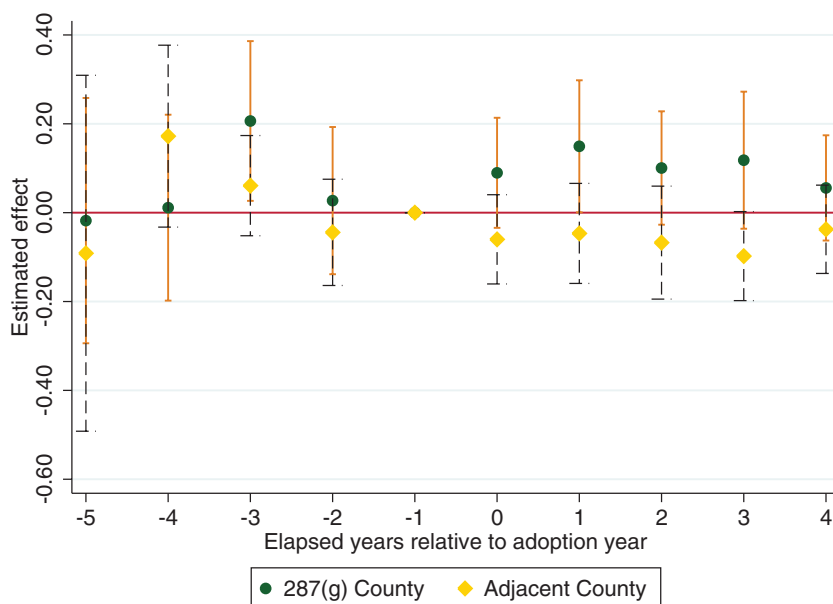


Figure 2. Estimated impact of 287(g) programs on log wages of farm workers for years before, during, and after adoption

on earnings. Pre-program estimates for program and adjacent counties follow each other closely, but in the post-law period they trend in opposite directions (and at about the same magnitude).

Taken together, these estimates show that program adoption changed the relative earning patterns across program and adjacent counties in a manner that is consistent with either a program effect or a program effect combined with a spillover effect of opposite sign in adjacents. However, without statistically different estimated paths relative to non-adjacent controls, it is not possible for us to choose between the two competing explanations. Additionally, we draw attention to immigrant presence estimates; they do not reveal that adjacent counties gain immigrants at a higher rate than non-adjacent controls.

County-Level Analysis

The ACS provides limited information on the U.S. agricultural sector. Therefore, we turn our attention to county-level data from the Census of Agriculture to further explore the effects of immigration laws on various agricultural outcomes at the county level.

The Census of Agriculture is conducted every five years, and as noted, all adoptions of 287(g) programs occurred after 2002. We thus estimate simple DD models using the

2002 wave of the survey (last wave prior to the first adoption) and the 2007 survey year (most recent survey wave available). We also use data from the 1997 survey wave to examine if changes in outcomes between 1997 and 2002 are statistically different in would-be adopters as a test of the parallel trends assumption.

DD estimates of program effects are presented in the upper panel of table 8. As before, the main specification includes every non-program county as a control. The specification also indicates post-program years in adopters and adjacents, as well as post-program years for counties located in 287(g) states, and those that experienced immigration raids between 2002 and 2007 (table 8).

In addition, we test pre-trends for each outcome variable to examine whether the treatment and the control group experienced similar paths before 287(g) enforcement went into effect. More specifically, we conduct t-tests to compare the 1997 to 2002 change in outcomes for would-be program counties to that in all non program counties and adjacent ones. The 1997 to 2002 changes in outcomes and p-values from the t-tests comparing these changes are reported in the lower panel of table 8.

We first examine the impact of 287(g) programs on hired labor as a share of total

Table 8. Impact of the 287(g) Program (All counties)

	Hired Labor Share	Expense per Worker	Workers per Acre	Share of Fuel Expenses	Log of Machinery Value	Share of Vegetable Acres	Log of Farm Income per Acre
Post-law	-.020*** (0.007)	534.127** (248.774)	-.328*** (0.057)	-.003 (0.006)	-.041 (0.082)	-.004* (0.002)	-.517*** (0.251)
Post-law adj	-.003 (0.004)	-461.588*** (154.963)	0.023 (0.03)	-.011*** (0.003)	0.02 (0.044)	-.0007 (0.001)	-.138* (0.133)
N	5,975	5,946	6,048	6,083	6,115	4,397	5,092
R ²	0.908	0.832	0.959	0.742	0.974	0.924	0.847
Areas	3,034	3,032	3,050	3,056	3,062	2,441	2,741
Tests on pre-trends							
DY _t = Y ₀₂ -Y ₉₇ (treatment counties)	0.024	1,359.1	-0.031	0.001	0.126	0.001	0.259
DY _n = Y ₀₂ -Y ₉₇ (all never-adopters)	0.004	863.1	-0.036	-0.008	-0.005	-0.001	-0.261
P-value (H ₀ : DY _t = DY _n)	0.011	0.43	0.946	0.0001	0.329	0.394	0.002
DY _a = Y ₀₂ -Y ₉₇ (adjacent counties)	-0.002	1,102.9	-0.092	0.0002	0.016	-0.004	-0.093
P-value (H ₀ : DY _t = DY _a)	0.007	0.729	0.659	0.784	0.425	0.034	0.183

production expenses. Results indicate that counties that enforced 287(g) programs experience a two percentage point drop compared to the control group. The share of hired labor in program counties was increasing before adoption, while in control (never-adopters) and adjacent counties it was either flat or slightly declining.

Between 2002 and 2007, the share of hired labor in program counties experienced a steeper decline than in either control or adjacent counties. Both the pre-program increase and post-program decline are statistically different ($p < 0.05$) from changes in never-adopters and adjacent counties. These results indicate that while hired labor expense shares were actually increasing disproportionately in 287(g) counties before the program, program adoptions may have led to a trend reversal and a decline in labor expense shares that is not matched by trends in never-adopters or adjacent non-program counties.

Similarly, we examine impacts on expense per hired worker, number of hired workers per acre of land in farms in 2002, fuel expense as a share of total expenses, value of

machinery per acre, share of vegetable acres, and farm income per acre.

Results indicate that expense per worker increased by \$534 in program areas relative to non-adjacent controls, while those in adjacent areas declined by \$462 ($p < 0.05$). These estimates are similar in direction and magnitude to the wage effects estimated with ACS data. The increase in expense per worker marks a 6.7% increase (from an average of \$8,007), while the point estimate of the change in log-earnings was 0.0672 (table 6).

This outcome, however, should be interpreted with caution, as information on how many weeks or hours per week each worker has worked during the year is not available. So this is not a wage rate; a higher expense per worker could reflect lower turnover (individuals being employed longer) or the same workers being employed for more hours each week, in addition to higher wage rates. However, a higher expense per individual worker is consistent with a tighter labor market.

Estimates also show that the number of workers hired decreased by one-third of a worker per acre, and the share of acres

planted with vegetables decreased by 0.4 percentage points.

The decrease in the number of workers per acre marks nearly a 16% change relative to an average of 2.1 workers. This outcome is also subject to the same cautions as above; because weeks worked and hours worked are not known, this could reflect similar use of labor-hours with less turnover.

We find no effect on the value of machines per acre, which runs counter to the expectation of mechanization in response to higher labor costs. However, machines could be rented instead of purchased. The estimated impact on the share of fuel expense (proxying for intensity of use of non-labor inputs) is statistically zero relative to all non-adjacent counties, but the change in the share of expenses that went to fuel was 8 percentage points higher in program counties than in adjacent counties. This is consistent with a relative increase in the intensity of machine use in program counties than in adjacent counties after the program.

Finally, we examine the impact of 287(g) programs on farm incomes. Farm income at the county level is the net cash farm income measured for all operations in the county for each year. Results indicate that farm income in adopting counties has decreased relative to the control group and relative to adjacents after the adoption of the law. The magnitudes, however, appear larger than expected (52% and 38% relative to all counties and adjacents, respectively).

Analysis of pre-trends presented in the bottom panel of table 8 indicate that program counties were on similar paths as controls and adjacents in terms of expense per worker and workers per acre. Pre-trends were also parallel between program and adjacent counties for the share of expenses that went to fuel (note that fuel share was only significantly different relative to adjacents after program adoption) and machine value per acre, as well as between program and control counties for the share of acres planted with vegetables.

Results also show a trend reversal in terms of the share of hired labor and share of vegetable acres (against adjacents only). Most importantly, none of the estimates appear to be the result of a differential trend continuation.

We also provide two alternative specifications in table A2 of the appendix. The first specification considers counties that have

more than \$1.2 million of agricultural crop sales. We do this since counties with very little agriculture may have a substantially different production function and may not be good controls. The second specification in the lower panel of table A2 includes total land in farms as a control, total land squared, total cropland in farms, and total cropland squared as additional controls. Given that farmers may change their crop production area as a response to potential changes in labor availability, this additional specification may offer additional insights on the extent to which observed effects on the outcome variable operate through an increased propensity to retire land in general or crop land in particular as labor costs increase. The results for both specifications are very similar to the main results.³

Discussion and Conclusions

We examine the impact of the 287(g) programs on the U.S. farming sector using individual data from the ACS, as well as county-level data from the Census of Agriculture.

The DD regressions highlight several effects of immigration laws on adopting jurisdictions. First, using individual level data, we find a decrease in the share of noncitizens in adopting counties in the post-program period. This finding provides evidence that, as many claim, 287(g) contracts enforced at the county level are causing declines in foreign-born populations that are larger than the plausible size of the violent-immigrant population the program was intended to target.

Second, we find that adopting counties have experienced a decrease in labor expenses as a share of total expenditures, a decrease in the number of workers hired, a decrease in farm incomes, and a decrease in vegetable acres compared to both non-adopters and adjacent counties. In addition, program counties seem to pay more per hired

³ In other specifications (available upon request), we have also estimated the ACS models with only counties adopting until 2007 (last year available in Census of Agriculture), and also estimated specifications with the Census of Agriculture while excluding counties that cannot be matched with PUMAs in the ACS. Results are robust to these sample exclusions, indicating that the sampling, geographic disclosure, and timing of each dataset are not driving our results.

worker in the post-law period and experience lower farm incomes. All this evidence is consistent with labor supply reductions being created by county-level 287(g) agreements.

Some of our findings are also consistent with a spillover effect that affects adjacent counties disproportionately. Specifically, the relative increase of farm worker earnings in program counties and a decline in adjacent counties can be interpreted as evidence of a program and a spillover effect, but neither earnings path alone was statistically different than that in non-adjacent control counties. Similarly, expense per worker (measured in the Census of Agriculture) in program areas had increased and that of adjacent areas had decreased by about the same percentage as earnings (in the ACS). These coefficients are statistically significant and can be interpreted as evidence of a program and a spillover effect.

However, none of the other results point clearly to the possibility of spillovers. Notably, immigrant presence regressions do not show disproportionate gains in immigrant populations in adjacent areas.

Our findings on the effects of immigration policy on wages and the number of illegal immigrants are similar to the findings of Devadoss and Luckstead (2011), who simulated the effects of a 10% increase in the domestic budget for enforcing policies (including employer fines for hiring illegal workers, an employee verification system for employers, and an increased number of raids) on unauthorized workers, and an increase (in probability of getting caught from 0.3 to 0.6) in border enforcement efforts from 1994 to 2007. Our analysis using ACS data suggests that a 20% reduction (relative to the national average) of illegal immigrants generates an increase of 6-7% in agricultural wages. The Devadoss and Luckstead (2011) analysis on the 2000-2007 period suggests that due to increased domestic enforcement, there is an average drop of 3.2% in the number of illegal immigrants, and an associated increase of 1.3% in agricultural wages. Thus, a 20% reduction in the number of illegal immigrants would generate an 8.1% increase in the wages of legal agricultural workers, which is similar to our findings.

Overall, this study provides some evidence that 287(g) programs have affected immigrant presence, agricultural labor prices, and use by farmers. Given the importance of labor in the agricultural sector, further

research using individual farm-level data may provide additional insights on the effects of immigration laws on the farming sector.

A wave of immigration laws were enacted by several states (e.g., Arizona and Georgia) in 2012 and 2013, and a better understanding of the effects of these localized efforts could be very useful for actors in the U.S. agricultural sector. This is particularly the case since, as Pham and Van (2010) suggest, the recent trend of sub-federal-level enforcement is new territory for immigration reform in the United States, with significant legal and policy implications. Previous waves of reform implemented federally (e.g., the 1986 IRCA) included measures that restrict undocumented labor use, but at the same time implemented other counter-measures designed to protect farms' labor supply (e.g., H-2A). Localized enforcement is likely to only put downward pressure on immigrant populations, and has the potential to significantly affect farm operations.

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Appendix

Table A1. 287(g) Contracts Signed

Jurisdiction	Law Enforcement Agency	Date
State-level enforcements		
Florida	Florida Department of Law Enforcement	2-Jul-02
Alabama	Alabama Department of Public Safety	10-Sep-03
Arizona	Arizona Department of Corrections	16-Sep-05
Arizona	Arizona Department of Public Safety	15-Apr-07
Colorado	Colorado Department of Public Safety	29-Mar-07
Georgia	Georgia Department of Public Safety	27-Jul-07
New Mexico	New Mexico Department of Corrections	17-Sep-07
Tennessee	Tennessee Highway Patrol / Department of Safety	25-Jun-08
Missouri	Missouri State Highway Patrol	25-Jun-08
Minnesota	Minnesota Department of Public Safety	22-Sep-08
Delaware	Delaware Department of Corrections	15-Oct-09
County-level enforcements		
California	San Bernardino County Sheriff's Office	19-Nov-05
California	Los Angeles County Sheriff's Office	1-Feb-05
California	Orange County Sheriff's Office	2-Nov-06
California	Riverside County Sheriff's Office	28-Apr-06
Colorado	El Paso County Sheriff's Office	17-May-07
Florida	Collier County Sheriff's Office	6-Aug-07
Florida	Bay County Sheriff's Office	15-Jun-08
Florida	Jacksonville Sheriff's Office	8-Jul-08
Georgia	Cobb County Sheriff's Office	13-Feb-07
Georgia	Whitfield County Sheriff's Office	4-Feb-08
Georgia	Hall County Sheriff's Office	29-Feb-08
Georgia	Gwinnett County Sheriff's Office	15-Oct-09
North Carolina	Mecklenburg County Sheriff's Office	27-Feb-06
North Carolina	Alamance County Sheriff's Office	10-Jan-07
North Carolina	Cabarrus County Sheriff's Office	2-Aug-07
North Carolina	Gaston County Sheriff's Office	22-Feb-07
North Carolina	Henderson County Sheriff's Office	25-Jun-08
North Carolina	Wake County Sheriff's Office	25-Jun-08
North Carolina	City of Durham Police Department	1-Feb-08
Oklahoma	Tulsa County Sheriff's Office	6-Aug-07
Arkansas	Benton County Sheriff's Office	26-Sep-07
Arkansas	City of Springdale Police Department	26-Sep-07
Arkansas	Rogers Police Department	25-Sep-07
Arkansas	Washington County Sheriff's Office	26-Sep-07
South Carolina	York County Sheriff's Office	16-Oct-07
South Carolina	Beaufort County Sheriff's Office	25-Jun-08
South Carolina	Charleston County Sheriff's Office	9-Nov-09
South Carolina	Lexington County Sheriff's Office	19-Aug-10
Tennessee	Davidson County Sheriff's Office	21-Feb-07
Virginia	Herndon Police Department	21-Mar-07
Virginia	Prince William-Manassas Regional Jail	9-Jul-07
Virginia	Rockingham County Sheriff's Office	25-Apr-07
Virginia	Shenandoah County Sheriff's Office	10-May-07
Virginia	Prince William County Police Department	26-Feb-08
Virginia	Prince William County Sheriff's Office	26-Feb-08
Virginia	Manassas Park Police Department	10-Mar-08
Virginia	Manassas Police Department	5-Mar-08
Virginia	Loudoun County Sheriff's Office	25-Jun-08
Ohio	Butler County Sheriff's Office	5-Feb-08
Maryland	Frederick County Sheriff's Office	6-Feb-08
Arizona	City of Phoenix Police Department	10-Mar-08
Arizona	Pima County Sheriff's Office	10-Mar-08
Arizona	Yavapai County Sheriff's Office	10-Mar-08

continued.

Table A1. continued.

Jurisdiction	Law Enforcement Agency	Date
Arizona	Maricopa county	14-Mar-08
Arizona	Florence Police Department	21-Oct-09
Arizona	City of Mesa Police Department	19-Nov-09
Texas	Farmers Branch Police Department	8-Jul-08
Texas	Harris County Sheriff's Office	20-Jul-08
Texas	Carrollton Police Department	12-Aug-08
New Jersey	Hudson County Department of Corrections	11-Aug-08
New Jersey	Monmouth County Sheriff's Office	15-Oct-09
Nevada	Las Vegas Metropolitan Police Department	8-Sep-08
Utah	Washington County Sheriff Office	22-Sep-08
Utah	Weber County Sheriff's Office	22-Sep-08
Connecticut	City of Danbury Police Department	15-Oct-09

Source: <http://www.ice.gov/news/library/factsheets/287g.htm>.

Table A2. Impact of the 287(g) Program (Additional Specifications)

	Hired Labor Share	Expense per Worker	Workers per Acre	Share of Fuel Expenses	Log of Machinery Value	Share of Vegetable Acres	Log of Farm Income per Acre
Counties with more than \$1.2 million of agricultural products							
Post-law	-.020*** (0.007)	519.029** (263.191)	-.324*** (0.059)	-.003 (0.005)	-.021 (0.064)	-.004** (0.002)	-.527*** (0.254)
Post-law adj	-.004 (0.004)	-473.685*** (165.422)	0.024 (0.032)	-.009*** (0.003)	0.038 (0.035)	-.0007 (0.001)	-.128 (0.143)
N	5,283	5,275	5,355	5,358	5,386	3,987	4,714
R ²	0.917	0.828	0.961	0.8	0.981	0.922	0.842
Areas	2,753	2,752	2,776	2,782	2,788	2,238	2,555
All counties with controls for land							
Post-law	-.022*** (0.007)	393.029 (256.318)	-.320*** (0.057)	-.001 (0.006)	0.062 (0.078)	-.003 (0.002)	-.519** (0.251)
Post-law adj	-.004 (0.004)	-585.093*** (168.274)	0.043 (0.03)	-.012*** (0.003)	-.018 (0.042)	-.002* (0.001)	-.131 (0.138)
Land in 100k Acres	0.001 (0.001)	-337.977*** (113.537)	0.021** (0.009)	0.005*** (0.0009)	0.19*** (0.013)	0.001*** (0.0003)	0.131*** (0.053)
Land squared	-.00006** (0.00003)	7.229** (2.967)	-.0005** (0.0002)	-.0002*** (0.00002)	-.005*** (0.0003)	-1.00e-05** (5.81e-06)	-.002 (0.001)
Cropland in 100k acres	-.019*** (0.003)	426.411** (208.326)	-.063*** (0.022)	-.002 (0.002)	-.060** (0.03)	0.002*** (0.0005)	-.293*** (0.0993)
Cropland squared	0.002*** (0.0003)	-57.441*** (14.065)	0.015*** (0.002)	0.0006*** (0.0002)	0.008*** (0.003)	-.0001* (0.00007)	0.015*** (0.009)
N	5,975	5,943	6,048	6,083	6,115	6,474	5,092
R ²	0.91	0.835	0.961	0.75	0.977	0.907	0.85
Areas	3,034	3,030	3,050	3,056	3,062	2,516	2,741